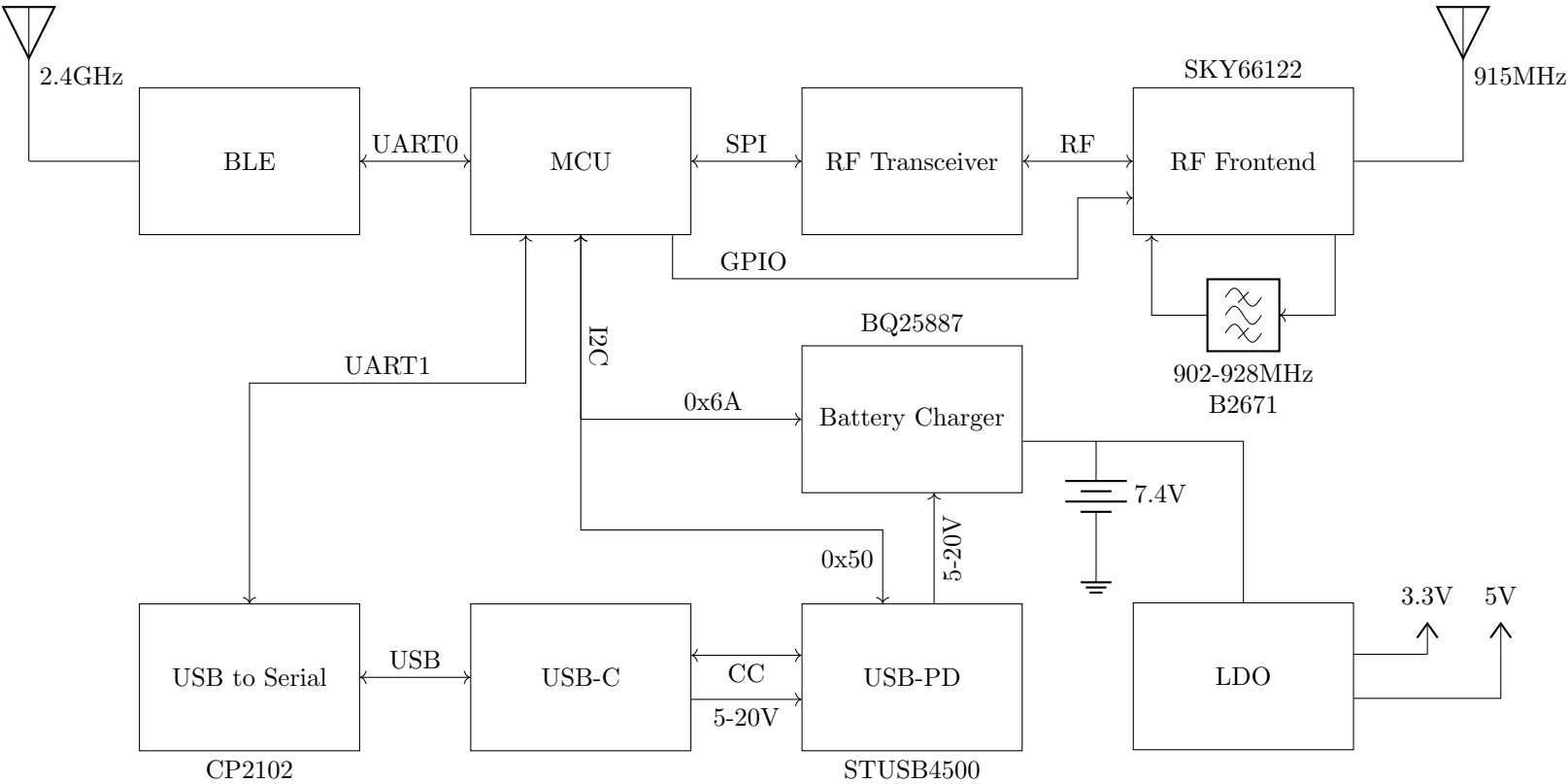
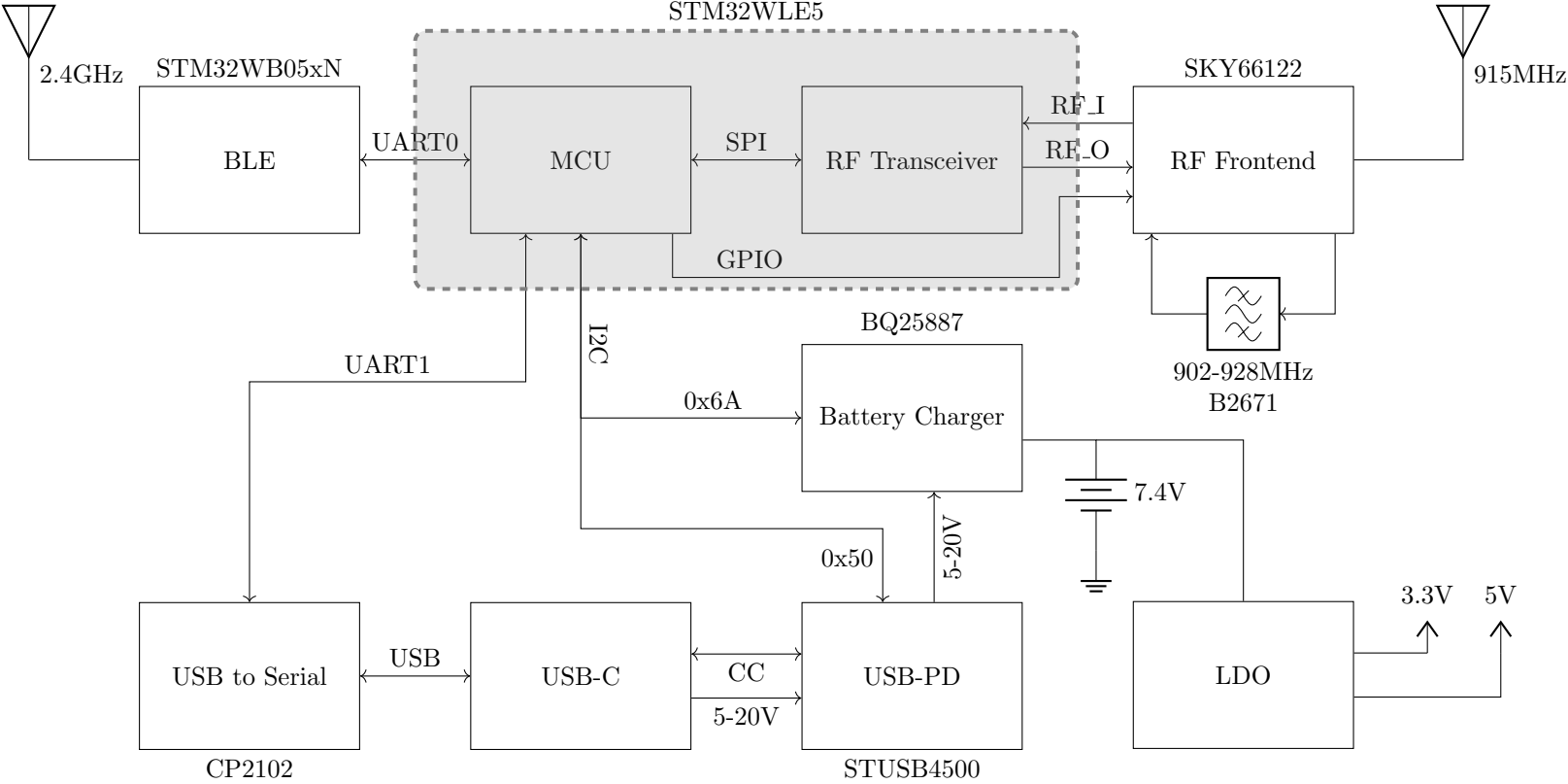


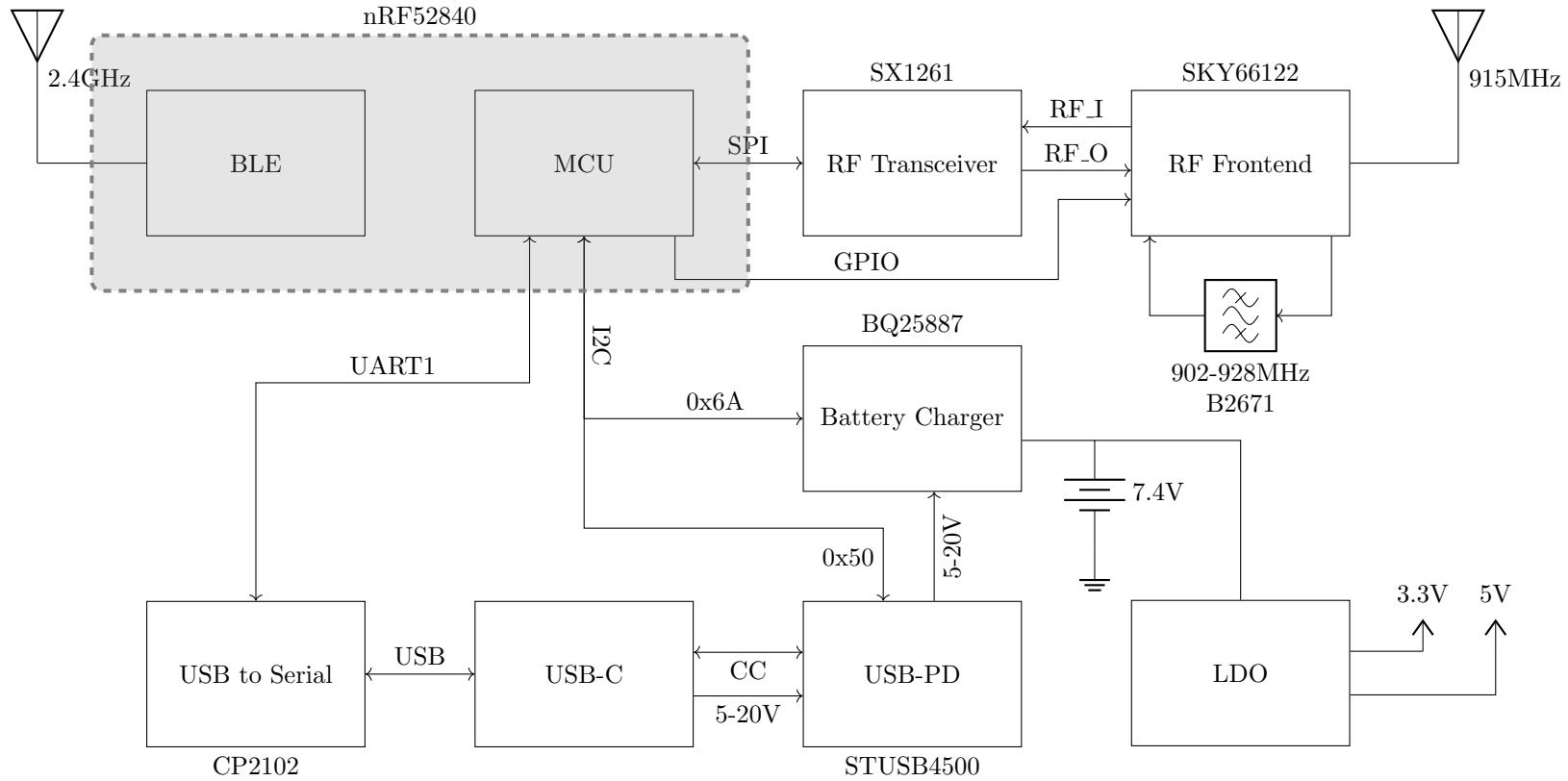
General Overview Block Diagram



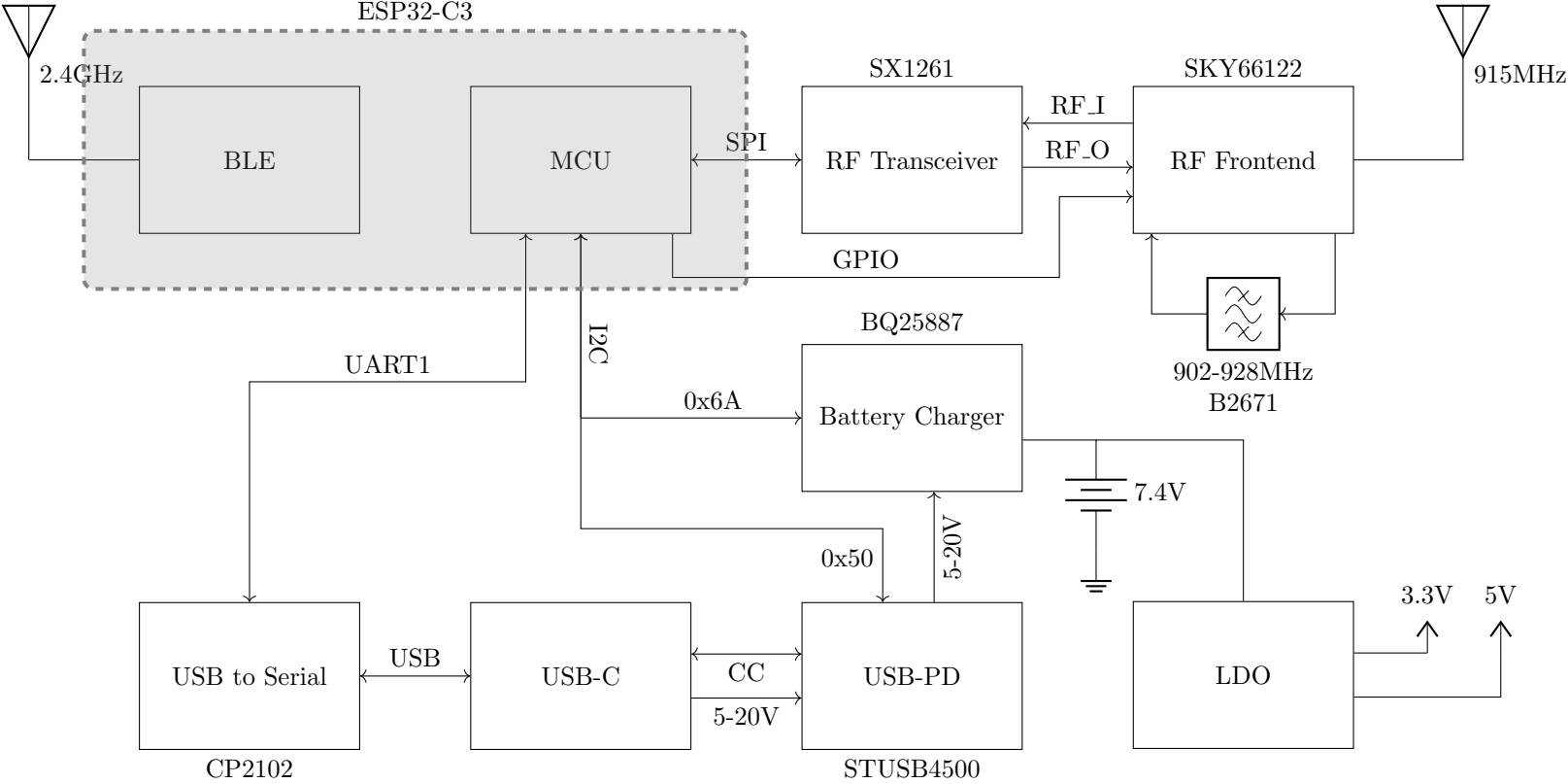
Component Selection Option A (STM32)



Component Selection Option B (nRF52840)

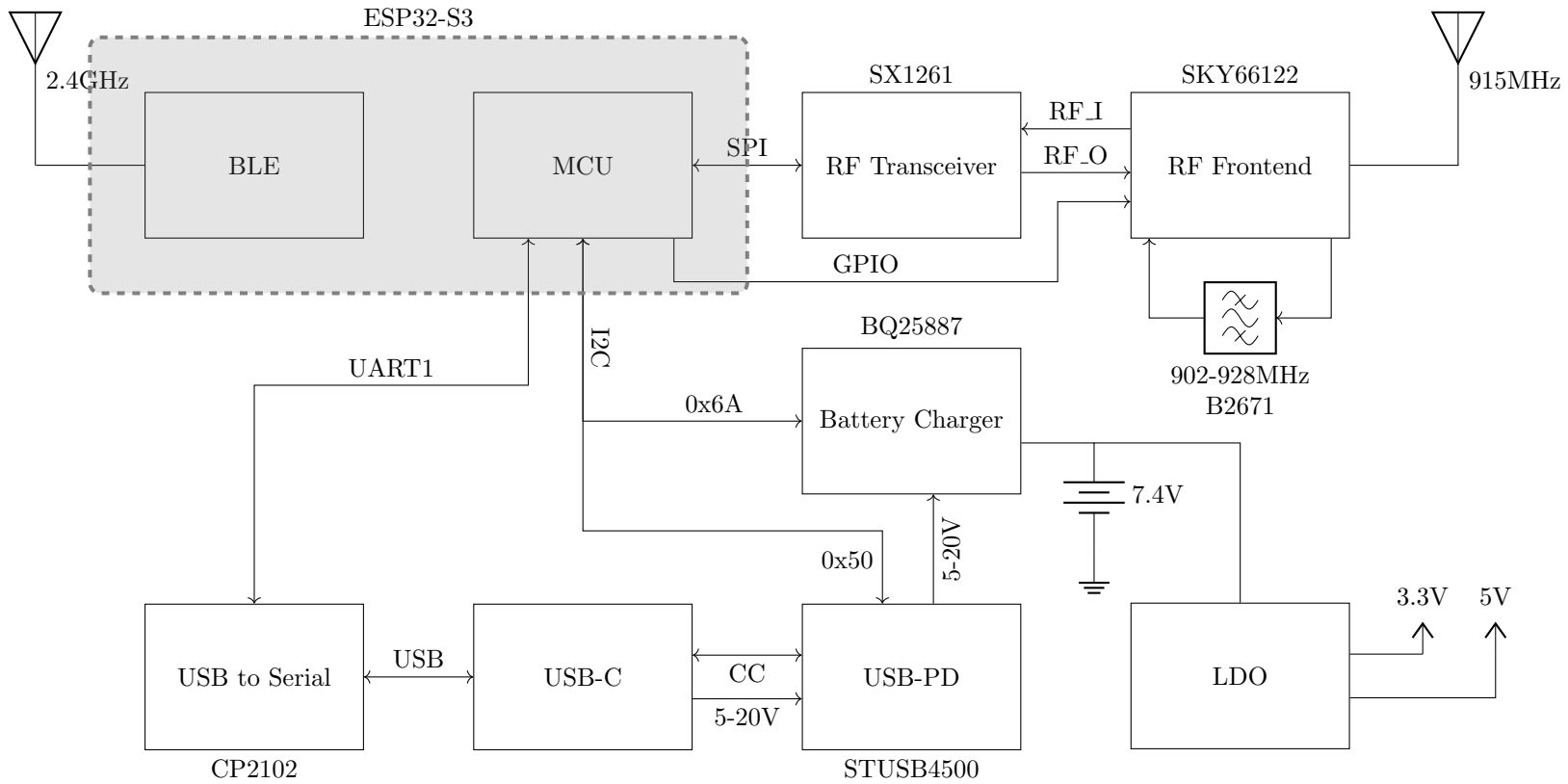


Component Selection Option C-1 (ESP32C3)



Component Selection Option C-2 (ESP32S3)

5



1 Power Consumption Analysis

1.1 Component Current Requirements

Option A

Component	Part Number	Peak Current (mA)	Continuous Current (mA)
BLE	STM32WB05xN	100	25
MCU	STM32WLE5	100	25
RF Transceiver		95	15
RF Frontend	SKY66122	500	50
Battery Charger	BQ25887	200	5
USB-PD	STUSB4500	30	8
USB-C	Generic	10	2
USB to Serial	CP2102	80	20
Total		150 mA	150 mA

2 LDO Selection